All Tasks Scheduling Orders (hard)

Problem Statement

There are ‘N’ tasks, labeled from ‘0’ to ‘N-1’. Each task can have some prerequisite tasks which need to be completed before it can be scheduled. Given the number of tasks and a list of prerequisite pairs, write a method to print all possible ordering of tasks meeting all prerequisites.

Example 1:

```
Input: Tasks=3, Prerequisites=[0, 1], [1, 2]
Output: [0, 1, 2]
Explanation: There is only possible ordering of the tasks.
```

Example 2:

```
Input: Tasks=4, Prerequisites=[3, 2], [3, 0], [2, 0], [2, 1]
Output:
1) [3, 2, 0, 1]
2) [3, 2, 1, 0]
Explanation: There are two possible orderings of the tasks meeting all
prerequisites.
```

Example 3:

```
Input: Tasks=6, Prerequisites=[2, 5], [0, 5], [0, 4], [1, 4], [3, 2],
[1, 3]
Output:
1) [0, 1, 4, 3, 2, 5]
2) [0, 1, 3, 4, 2, 5]
3) [0, 1, 3, 2, 4, 5]
4) [0, 1, 3, 2, 5, 4]
5) [1, 0, 3, 4, 2, 5]
6) [1, 0, 3, 2, 4, 5]
7) [1, 0, 3, 2, 5, 4]
8) [1, 0, 4, 3, 2, 5]
9) [1, 3, 0, 2, 4, 5]
10) [1, 3, 0, 2, 5, 4]
11) [1, 3, 0, 4, 2, 5]
12) [1, 3, 2, 0, 5, 4]
13) [1, 3, 2, 0, 4, 5]
```

Try it yourself

Try solving this question here:

Java

```java
import java.util.*;

class Solution {

  List<List<Integer>> orders;

  public Solution() {
    orders = new ArrayList<>();
  }

  public List<List<Integer>> printOrders(int tasks, int[][] prerequisites) {
    // TODO: Write your code here

    return orders;
  }
}
```

Test Case | Run Code Result | Stdout

```
3
[[0,1],[1,2]]
```

Use Example Testcases | Run | Submit

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Solution: All Tasks Scheduling Orders

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